

Albany County Behavioral EV Charging Results

Full dataset run with ZIP/ZCTA adoption scaling, H3-8 charging geography, and
GridUp-style charger availability

NYSERDA EV Adoption and Charging Model

June 23, 2026

This week's status

Area	Current status
Model geography	Adoption now scales from home ZIP/ZCTA; charging load is placed at destination H3.
Charging behavior	Availability now follows the GridUp-style inverse-demand choice model; allocation still uses conditional expected value across ranked stops.
Managed charging	Applied only to managed-eligible home/work L2 sessions; public, civic, and DCFC sessions remain unmanaged.
Full run	Full Albany dataset completed with H3 required and ZIP/ZCTA adoption scaling enabled.
Visualization	Albany ZIP adoption, H3 charging-demand, and selected-H3 point maps rebuilt from the current forecast/model outputs.

Remaining trip-chain modeling decision: final stops

Final activity distribution for chains that do not end at home

Final activity type	Chains
Social / Recreational	7,572
Work	3,300
School / Daycare / Religious	1,620
Shopping / Errands	909
Other	715
Transport / drop-off / pick-up	450
Meals	117
Medical / Dental	105

- ▶ **As observed:** final non-home stops remain non-home overnight stops.
- ▶ **Completed chain:** add a synthetic return-home leg and overnight home dwell.

Geography design: home adoption vs. physical load

Core decision

Keep home adoption geography separate from charging-load geography.

Use case	Target geography	Reason
EV adoption forecast	Home ZIP/ZCTA	Adoption model is ZIP-level; join from home coordinate to ZIP/ZCTA.
Physical charging load	Destination H3	Load appears where vehicles charge, not necessarily where they live.

GridUp-style availability and charging allocation

Availability model

Configured probabilities are interpreted as peak stopped-vehicle-hour availability, then adjusted by local stopped-vehicle demand:

$$q_{j,t} = \min \left(1, q_j^{\text{peak}} \frac{N_{\ell(j)}^{\text{peak}}}{N_{\ell(j),t}} \right)$$

where $\ell(j)$ is charger assumption, charger type, and rounded destination point.

Allocation method

Carry probability states across ranked charging opportunities:

$$\mathcal{S}_j = \{(p_k, r_k) : \text{probability } p_k, \text{ remaining energy } r_k\}$$

At each stop, split every state into available and unavailable branches, allocate:

$$E_{j,k} = \min(r_k, C_j)$$

and aggregate expected event energy:

$$\mathbb{E}[E_j] = \sum_k p_k \cdot q_j \cdot E_{j,k}$$

Stop ranking is home, work, long public, civic, then quick public.

Managed vs. unmanaged charging

Unmanaged

Charging begins at arrival and uses charger power until allocated energy is delivered or the dwell window closes:

$$T_s = \min \left(\tau_s, \frac{E_s}{P_s} \right)$$

Managed

Only managed-eligible charger types are spread across the dwell window:

$$\bar{P}_s = \min \left(P_s, \frac{E_s}{\tau_s} \right)$$

- ▶ Peak-hour availability assumptions: single-family home L2 1.00, multifamily home L2 0.50, work L2 0.50, civic L2 0.50, long public L2 1.00, quick public DCFC 0.35.
- ▶ GridUp adjustment raises availability in lower-demand stopped-vehicle hours and clips probability at 1.00.
- ▶ This keeps home/work management separate from public/DCFC charging behavior.

Adoption scaling logic

Forecast scaling equation

$$L_{g,t,c,y} = \sum_h L_{h,g,t,c}^{\text{fullEV}} \cdot A_{h,y,s} \cdot G_{h,y,s}$$

Symbol	Meaning
h	Home ZIP/ZCTA used for adoption attribution.
g	Charging H3 cell where load physically appears.
c	Charger category / charging location type.
$A_{h,y,s}$	Adoption fraction for home geography, year, and scenario.
$G_{h,y,s}$	Vehicle growth factor for home geography, year, and scenario.

Result Summary

Question	Current result
Full-EV daily energy delivered	3.723 GWh per modeled day under both managed and unmanaged modes.
Full-EV peak load	337.3 MW unmanaged at 20:00; 229.8 MW managed at 03:00.
2027 baseline adoption load	174.6 MWh per modeled day; 15.81 MW unmanaged peak.
Managed charging effect	About 31.9% lower full-EV peak and 31.9% lower 2027 baseline peak.

Takeaway: The model now produces a home-ZCTA by charging-H3 by time load surface, then scales it with the ZIP-level adoption forecast.

Modeling Geography

Adoption scaling geography

- ▶ Adoption forecast is ZIP/ZCTA-level.
- ▶ Each modeled vehicle day keeps its home ZCTA.
- ▶ Scaling is applied by `home_zcta`, forecast year, and adoption scenario.

Charging-load geography

- ▶ Charging load is placed at the destination charging location.
- ▶ Primary planning geography is H3 at resolution 8.

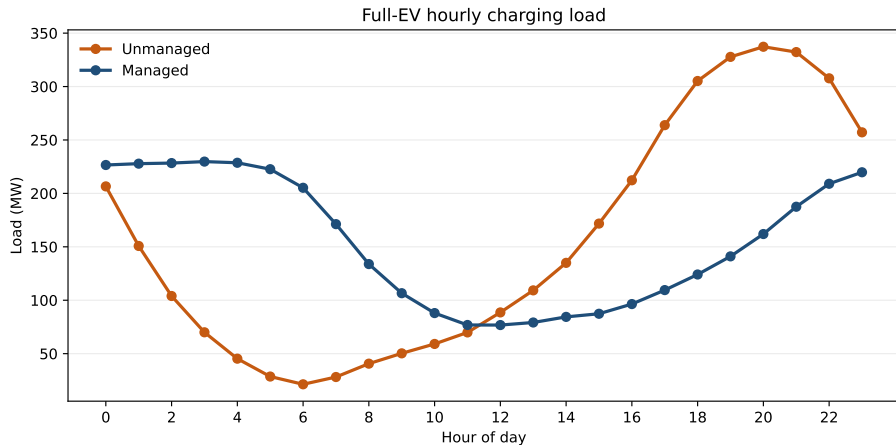
Full-EV Energy Accounting

Metric	Value per modeled day
Vehicle days	216,960
Total network miles	8.964 million miles
Mean daily network miles	41.3 miles per vehicle day
Daily travel energy need	3.765 GWh
Expected charging energy delivered	3.723 GWh
Unmet energy	0.042 GWh
Unmet share of daily travel energy	1.12%

Interpretation

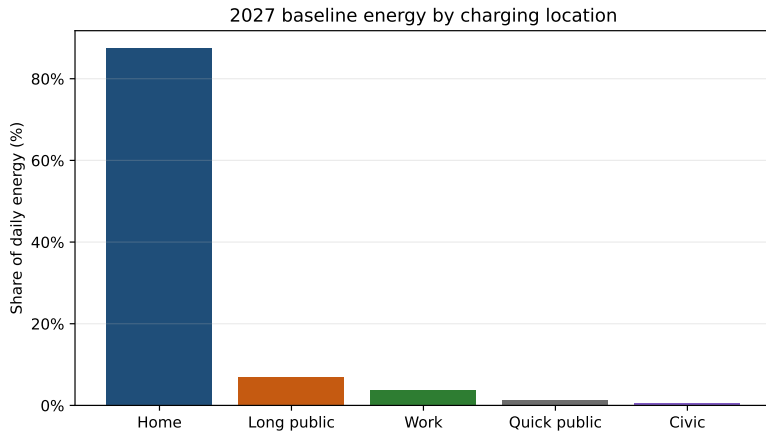
The unmet-energy metric is not a grid load. It is a behavioral feasibility signal showing where the current stop, dwell, charger-access, and power assumptions cannot fully replenish modeled daily travel energy.

Full-EV Hourly Load Shape



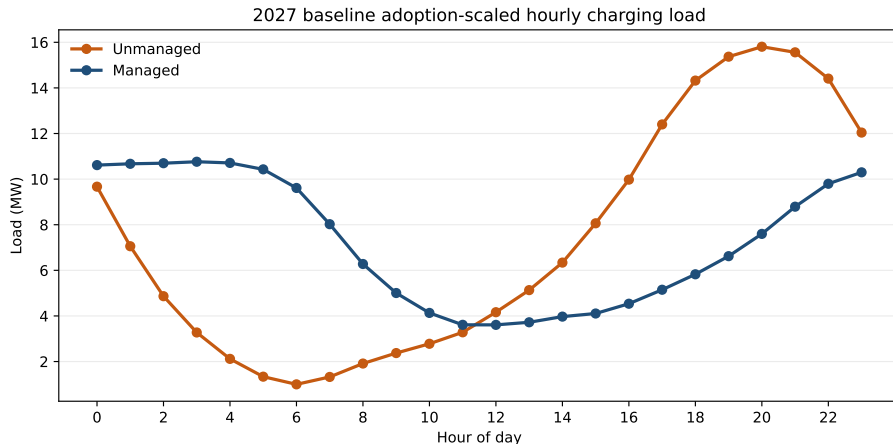
Takeaway: Managed charging shifts the aggregate peak from 20:00 to 03:00 and reduces the full-EV peak from 337.3 MW to 229.8 MW.

Charging Energy Is Concentrated at Home



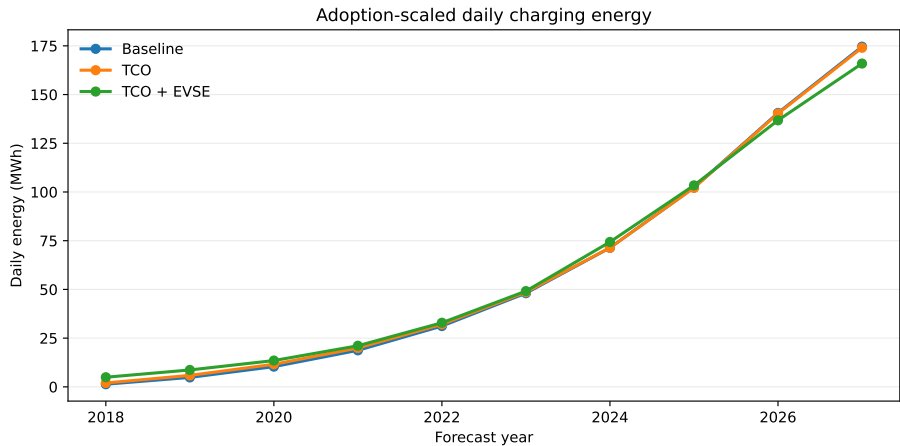
Takeaway: Home charging accounts for 87.4% of 2027 baseline daily energy; long public contributes 7.1%, work 3.8%, quick public 1.3%, and civic 0.5%.

2027 Baseline Adoption-Scaled Load

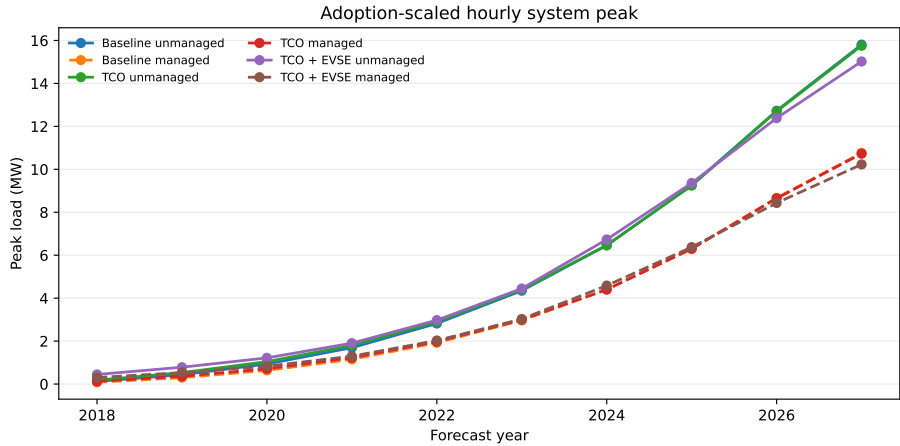


Takeaway: For 2027 baseline adoption, managed charging lowers the county-wide hourly peak from 15.81 MW to 10.76 MW, with the same total daily energy of 174.6 MWh.

Adoption-Scaled Energy by Forecast Year



Adoption-Scaled Peak by Forecast Year

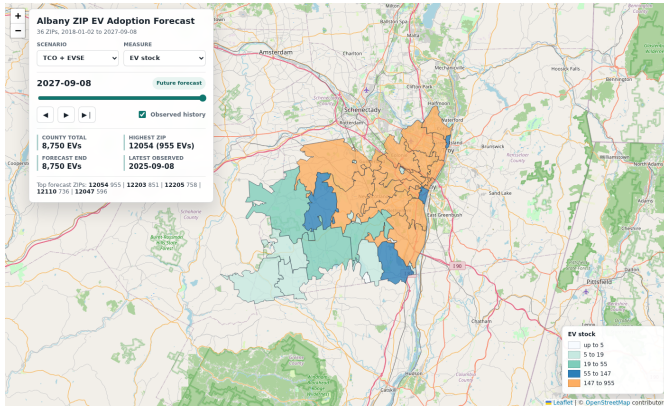


Interactive Visualization Outputs

Visualization	Purpose
ZIP adoption forecast map	Shows forecast EV stock or adoption share by home ZIP/ZCTA, scenario, and date. This is the adoption side of the workflow.
H3 charging-demand map	Shows adoption-scaled charging demand by destination H3 cell, scenario, forecast year, managed mode, metric, and hour. This is the physical-load side of the workflow.
Selected-H3 point map	Drills into one H3 cell and shows the destination points that create load inside that cell. This supports QA and future feeder or transformer joins.

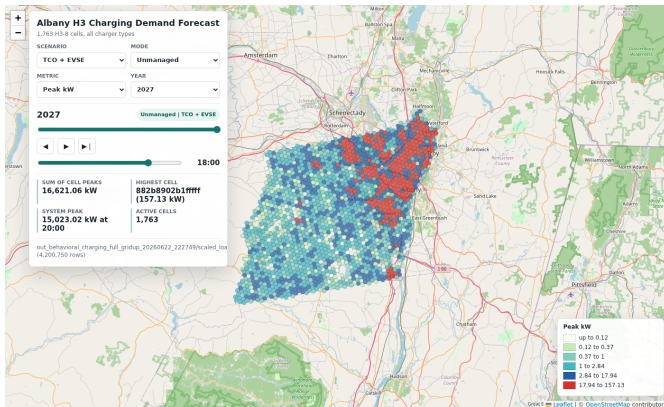
Files: `visualization/albany_zip_adoption_forecast_map.html`, `visualization/albany_h3_charging_demand_forecast_map.html`, and selected-H3 point map/CSV outputs.

Visualization: Adoption Forecast



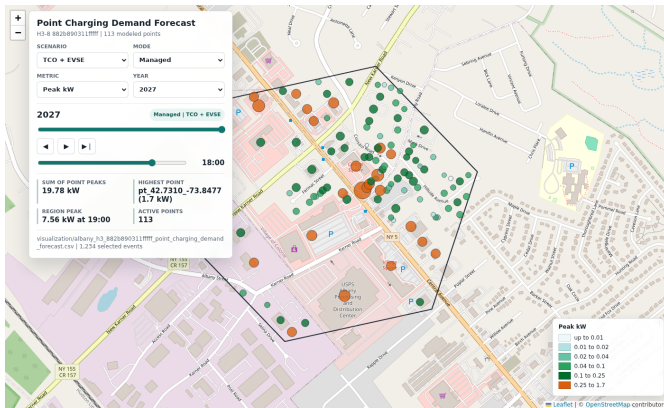
Takeaway: This view is for checking the home-side adoption forecast before it is used to scale the behavioral charging surface.

Visualization: H3 Charging Demand Forecast



Takeaway: This view converts the modeled behavior into spatial grid demand: scenario, year, managed mode, metric, and hour can be inspected at H3-8 resolution.

Visualization: Point-Level Demand Within One H3 Cell



Takeaway: The point-level view explains which destination points are driving load inside a selected H3 cell, which is useful before linking demand to utility assets.

2027 Baseline H3 Hotspots

Unmanaged top H3-hours

H3 cell	Hour	kW
882b8902b1fffff	20	165.6
882b8902b1fffff	19	163.2
882b890025fffff	19	160.6
882b8902b1fffff	18	159.9
882b890025fffff	18	156.4

Managed top H3-hours

H3 cell	Hour	kW
882b8902b1fffff	3	107.4
882b8902b1fffff	1	107.1
882b8902b1fffff	0	106.6
882b8902b1fffff	4	106.6
882b8902b1fffff	2	106.6

Current Caveats

- ▶ Albany trip data supports destination coordinates, purpose, and dwell time, but not detailed destination land use.
- ▶ Office vs. non-office workplace charging and DCFC 150/250/350 kW splits are still approximated with configurable assumptions.
- ▶ Charger-access and charger-power assumptions are configurable but not yet calibrated against local charger inventory and housing data.
- ▶ The adoption forecast in this run covers 2018–2027, so longer-term grid planning requires a longer adoption horizon.

Next Steps

1. Map charging H3 cells to feeders, substations, or utility service polygons when those layers become available.
2. Join destinations to land-use and EVSE inventory layers to calibrate workplace and DCFC charger choices.
3. Calibrate synthetic vehicle-day weights to observed Albany County vehicle stock and EV registrations.
4. Extend the ZIP/ZCTA adoption forecast and expand daily profiles to full-year load shapes for planning studies.
5. Run sensitivity cases for charger access, managed eligibility, DCFC mix, and energy efficiency.